

ASM Temporal Modeling: Final Alternative Recommendations

Date: 2026-04-15 **Contract:** #4 of 4 — Research Synthesis **Synthesized from:**

- [popw-better-alternatives-final.md](#)
- [asm-alternatives-sota-research.md](#)
- [asm-alternatives-dataset-comparison.md](#)

Constraints acknowledged:

- **POPW system:** IKEA ASM (254 videos, 33 classes, 685K frames), RTX 3060 GPU
- **Evaluation metric:** Frame-level accuracy at stride=1
- **POPW current:** 95.2% frame accuracy, ~24M params, ~15 GFLOPs

Executive Summary

POPW (95.2% IKEA ASM) already exceeds all surveyed alternatives in raw accuracy. However, two alternatives offer meaningful advantages in efficiency or architectural compatibility: **PoseC3D** (best-in-class accuracy/efficiency tradeoff for skeleton-based ASM) and **Perceiver IO** (native pose-conditioned cross-attention with linear complexity). A third method, **3DINAction**, provides a valid point-cloud modality alternative but with lower accuracy.

The critical research finding is a **benchmark gap**: MS-TCN++ and ASFormer — the canonical temporal action segmentation (TAS) methods — have never been evaluated on IKEA ASM, Assembly101, or IndustReal. Their SOTA claims on 50 Salads and Breakfast are not directly applicable to assembly domain.

Comparison Table: Methods with Assembly Dataset Benchmarks

Method	Year	Params (M)	GFLOPs	IKEA ASM	IndustReal	Assembly101	RTX 3060
POPW (this work)	2026	~24.0	~15	95.2%	N/E	N/E	✓
PoseC3D (Duan et al.)	2022	~33.8	~39	73.8%	N/E	N/E	✓
P3D-C (Qiu et al.)	2017	27.6	58	60.46%	N/A	N/A	✓
I3D (Carreira & Zisserman)	2017	25.0	~50	57.57%	N/A	N/A	⚠ (flow expensive)

Method	Year	Params (M)	GFLOPs	IKEA ASM	IndustReal	Assembly101	RTX 3060
3DINAction							
(Ben-Shabat et al.)	2023	~3.5 est.	N/A	52.91%	N/A	N/A	N/A
Non-local Networks							
(Wang et al.)	2018	+0.8M/block	8.2	N/E	N/E	N/E	✓
Perceiver IO							
(Jaegle et al.)	2022	44M (15.2M light)	78.5 (3.4 light)	N/E	N/E	N/E	✓
MS-TCN++							
(Li et al.)	2021	10.4	N/A	N/E	N/E	N/E	N/E
ASFormer							
(Yi et al.)	2021	6.8	N/A	N/E	N/E	N/E	N/E

Legend: N/E = Not Evaluated on this dataset; N/A = Not Available in literature

Alternative #1: PoseC3D — Skeleton-Based ASM Baseline

Why PoseC3D Is a Valid Better Alternative

PoseC3D (Duan et al., CVPR 2022) is the **best-performing skeleton-based method** on IKEA ASM at 73.8% frame accuracy, using 3D CNNs on pose heatmap volumes rather than graph convolutions. It outperforms all non-POPW RGB alternatives on the exact same dataset.

Key advantages over P3D-C and I3D (RGB baselines):

- PoseC3D achieves **73.8%** vs P3D-C's 60.46% — a **13.4 point gap** on the same IKEA ASM split
- PoseC3D uses ~39 GFLOPs vs P3D-C's 58 GFLOPs — **32% more efficient**
- Pose-based input is robust to visual occlusions common in assembly (hands covering parts)
- PoseC3D's skeleton modality is architecturally compatible with POPW's keypoint detection head

PoseC3D vs POPW:

- POPW (95.2%) substantially outperforms PoseC3D (73.8%) on accuracy (+21.4 points)
- POPW (15 GFLOPs) is more efficient than PoseC3D (39 GFLOPs) by ~2.6×
- POPW combines skeleton + RGB via FiLM conditioning; PoseC3D is skeleton-only
- **Conclusion:** PoseC3D validates the value of pose-based features (as POPW's FiLM conditioning suggests) but is not superior to POPW's combined approach

Key Numbers

Metric	Value
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Metric	Value
Parameters	~33.8M
GFLOPs	~39
IKEA ASM accuracy	73.8%
Backbone	C3D on heatmap volumes
Modality	Skeleton-only (poses + heatmaps)
RTX 3060 compatible	✓

Relevance to POPW

PoseC3D confirms that **pose-enhanced temporal modeling is the correct direction** for assembly understanding. POPW's FiLM-conditioned multi-task architecture extends this insight by fusing pose with RGB features rather than using pose alone — explaining POPW's 21.4-point accuracy advantage over PoseC3D.

Alternative #2: Perceiver IO — Temporal Head Replacement

Why Perceiver IO Is a Better Temporal Architecture

Perceiver IO (Jaegle et al., ICML 2022) is recommended in the POPW alternatives research as a **drop-in replacement for the Temporal Feature Bank**. Its core advantage is transforming retrieval from $O(n^2)$ quadratic complexity to $O(n \cdot d)$ linear complexity, while enabling **native pose-conditioned cross-attention**.

Architectural advantages:

- Cross-attention queries can be pose embeddings from POPW's keypoint head — directly implements "pose asks: which video features are relevant for this pose configuration?"
- Linear complexity $O(n \cdot d)$ vs Feature Bank's $O(n^2)$ — critical for long assembly sequences
- No gradient conflict with frozen ResNet-50 backbone (cross-attention trains independently)
- Lightweight variant (15.2M params, 3.4 GFLOPs) achieves 58.7% SSv2 — 98% of full Perceiver IO accuracy at 5% of the GFLOPs

Accuracy vs alternatives:

- Perceiver IO: 59.8% SSv2, 77.9% K400 (cross-attention retrieval paradigm)
- Non-local Networks: 61.2% SSv2, 76.5% K400 (pairwise self-attention paradigm)
- Both exceed P3D-C (SSv2 ~55%) and approach I3D on K400

Key Numbers

Metric	Full Perceiver IO	Lightweight Variant
Parameters	44M	15.2M
GFLOPs (T=8)	78.5	3.4

Metric	Full Perceiver IO	Lightweight Variant
SSv2 accuracy	59.8%	58.7%
K400 accuracy	77.9%	—
Complexity	O(n·d) linear	O(n·d) linear
Pose-conditioned	✓ Native	✓ Native
Drop-in (no backbone)	✓	✓
RTX 3060 compatible	✓	✓

Relevance to POPW

Perceiver IO solves the Feature Bank's quadratic scaling problem and adds native pose-query capability. For POPW's 254-video IKEA ASM dataset and RTX 3060 constraint, the **lightweight variant (15.2M params, 3.4 GFLOPs)** is the recommended configuration. This is compatible with POPW's FiLM architecture — pose embeddings from the keypoint head serve directly as Perceiver latent queries.

Alternative #3: 3DINAction — Point Cloud Baseline

3DINAction (Ben-Shabat et al., CVPR 2023) is the **only method** with IKEA ASM evaluation using depth/point cloud data. It achieves 52.91% frame accuracy using hierarchical t-patch modules on 3D point cloud sequences.

Why it ranks below PoseC3D and Perceiver IO:

- Lower accuracy (52.91%) than PoseC3D (73.8%) on the same dataset
- Params estimated ~3.5M (lowest of all alternatives) but no GFLOPs reported
- Point cloud modality requires RGB-D sensor input not in POPW's current pipeline
- No Assembly101 or IndustReal evaluation

Valid use case: When POPW is extended to multi-modal RGB-D input for occluded assembly scenarios, 3DINAction's t-patch concept (tracking local region motion through time) is architecturally similar to POPW's pose-aware temporal modeling.

Research Gap: TAS Methods Not Benchmarked on Assembly Datasets

[[research/asm-alternatives-sota-research|See SOTA research]] for full details on MS-TCN++ and ASFormer evaluation gaps.

Critical Finding

MS-TCN++ and ASFormer — the canonical temporal action segmentation methods — have NEVER been evaluated on IKEA ASM, Assembly101, or IndustReal.

Method	50 Salads	Breakfast	IKEA ASM	Assembly101	IndustReal
MS-TCN++	85.4%	65.0%	N/E	N/E	N/E

Method	50 Salads	Breakfast	IKEA ASM	Assembly101	IndustReal
ASFormer	86.7%	67.1%	N/E	N/E	N/E
POPW	N/E	N/E	95.2%	N/E	N/E

This creates a **critical cross-dataset evaluation gap**: POPW cannot be directly compared to MS-TCN++ or ASFormer because they operate on disjoint dataset sets.

Implications for POPW

1. POPW's 95.2% on IKEA ASM is NOT directly comparable to MS-TCN++ (85.4% on 50 Salads) — different datasets, different task difficulty
2. 50 Salads has 17 classes vs IKEA ASM's 33 classes — POPW's task is proportionally harder
3. **Future work**: Cross-dataset evaluation (train POPW on 50 Salads, compare to MS-TCN++) would enable fair comparison
4. Assembly101 and IndustReal have **zero published ASM-specific benchmarks** — POPW could establish the first baselines

Final Recommendations

For POPW's Current Deployment (IKEA ASM, 254 videos, RTX 3060)

POPW is already the best option at 95.2% IKEA ASM accuracy, 15 GFLOPs, and RTX 3060 compatibility. No surveyed alternative matches this accuracy/efficiency combination.

If Seeking Architectural Improvements Within POPW

1. **Replace Temporal Feature Bank with Perceiver IO (lightweight)**: Adds 15.2M params, 3.4 GFLOPs, enables native pose-conditioned cross-attention retrieval. Preserves POPW's multi-task design.
2. **Add Non-local Networks blocks as temporal head complement**: +0.8M/block, 8.2 GFLOPs (5 blocks), 61.2% SSv2. Provides global pairwise attention to complement POPW's existing temporal modeling.

If Seeking an Alternative Baseline for Comparison

- **Use PoseC3D** (73.8% IKEA ASM, 39 GFLOPs) as the strongest published skeleton-based alternative
- **Do NOT compare POPW to MS-TCN++ or ASFormer** without first running cross-dataset evaluation

Sources

- **POPW**: Internal work (2026) — 95.2% IKEA ASM, 24M params, 15 GFLOPs
- **PoseC3D**: Duan et al., CVPR 2022, arXiv:2204.13810 — 73.8% IKEA ASM, 33.8M params, 39 GFLOPs
- **P3D-C**: Qiu et al., CVPR 2017, arXiv:1711.10305 — 60.46% IKEA ASM, 27.6M params, 58 GFLOPs
- **I3D**: Carreira & Zisserman, CVPR 2017, arXiv:1705.07750 — 57.57% IKEA ASM, 25M params, ~50 GFLOPs
- **3DINAction**: Ben-Shabat et al., CVPR 2023, arXiv:2303.06346 — 52.91% IKEA ASM, ~3.5M est., N/A GFLOPs

- **Perceiver IO**: Jaegle et al., ICML 2022, arXiv:2107.14795 — 59.8% SSv2, 44M/15.2M params, 78.5/3.4 GFLOPs
 - **Non-local Networks**: Wang et al., CVPR 2018, arXiv:1711.07971 — 61.2% SSv2, +0.8M/block, 8.2 GFLOPs
 - **MS-TCN++**: Li et al., TPAMI 2021, arXiv:2006.09220 — 85.4% 50 Salads, 10.4M params
 - **ASFormer**: Yi et al., BMVC 2021, arXiv:2110.08568 — 86.7% 50 Salads, 6.8M params
 - **IKEA ASM Dataset**: Ben-Shabat & Kumar, ICCV 2021, <https://ikea.asm.work/>
 - **Assembly101**: Sener et al., CVPR 2022, <https://assembly-101.github.io/>
 - **IndustReal**: Schoonbeek et al., WACV 2024, <https://github.com/TimSchoonbeek/IndustReal>
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Research synthesized 2026-04-15 for POPW multi-task assembly understanding project